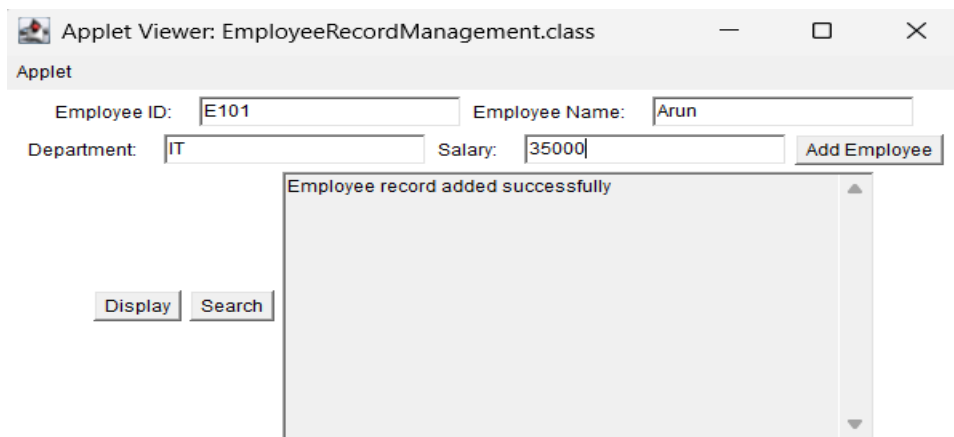
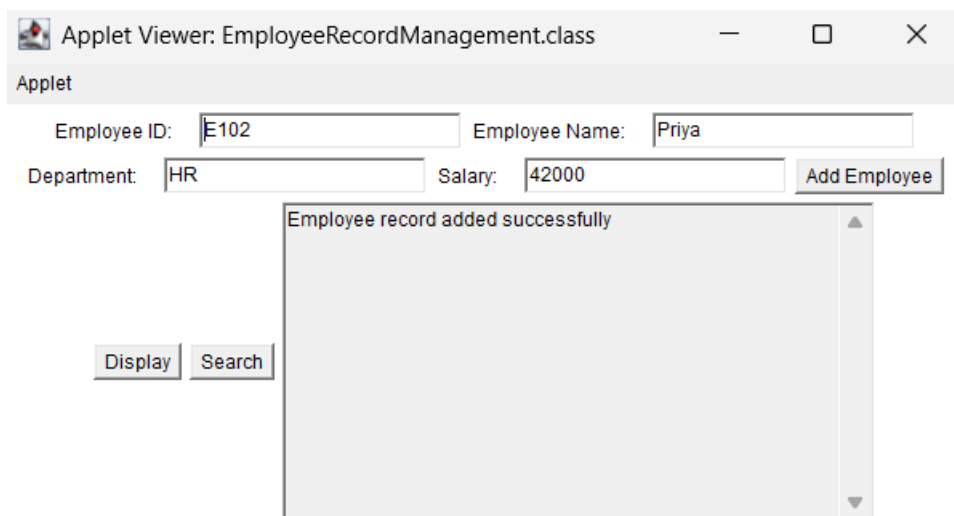




Applet started.



Applet started.

JAVA CODE:

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
import java.io.*;

public class EmployeeRecordManagement extends Applet
    implements ActionListener {

    Label idLabel, nameLabel, departmentLabel, salaryLabel;

    TextField idField, nameField, departmentField, salaryField;

    Button addButton, displayButton, searchButton;

    TextArea outputArea;

    String fileName = "employees.txt";

    public void init() {

        setLayout(new FlowLayout());

        idLabel = new Label("Employee ID:");
        nameLabel = new Label("Employee Name:");
        departmentLabel = new Label("Department:");
        salaryLabel = new Label("Salary:");

        idField = new TextField(20);
        nameField = new TextField(20);
        departmentField = new TextField(20);
        salaryField = new TextField(20);
```

```
        addButton = new Button("Add Employee");
        displayButton = new Button("Display");
        searchButton = new Button("Search");

        outputArea = new TextArea(12, 50);
        outputArea.setEditable(false);

        add(idLabel);
        add(idField);

        add(nameLabel);
        add(nameField);

        add(departmentLabel);
        add(departmentField);

        add(salaryLabel);
        add(salaryField);

        add(addButton);
        add(displayButton);
        add(searchButton);

        add(outputArea);

        addButton.addActionListener(this);
        displayButton.addActionListener(this);
        searchButton.addActionListener(this);
    }

    public void actionPerformed(ActionEvent e) {
```

```

        if (e.getSource() == addButton) {
            addEmployee();
        }

        else if (e.getSource() == displayButton) {
            displayEmployees();
        }

        else if (e.getSource() == searchButton) {
            searchEmployee();
        }
    }

    public void addEmployee() {

        String id = idField.getText();
        String name = nameField.getText();
        String department = departmentField.getText();
        String salary = salaryField.getText();

        if (id.equals("") || name.equals("")
            || department.equals("") || salary.equals("")) {

            outputArea.setText("Please enter all employee
details.");
            return;
        }

        try {

            FileWriter fileWriter =
                new FileWriter(fileName, true);

```

```
        BufferedWriter bufferedWriter =
            new BufferedWriter(fileWriter);

        bufferedWriter.write(
            id + "," +
            name + "," +
            department + "," +
            salary
        );

        bufferedWriter.newLine();

        bufferedWriter.close();

        outputArea.setText(
            "Employee record added successfully"
        );

        idField.setText("");
        nameField.setText("");
        departmentField.setText("");
        salaryField.setText("");
    }

    catch (IOException ex) {

        outputArea.setText(
            "Error: Unable to write employee record."
        );
    }
}
```

```
}
```

```
public void displayEmployees() {
```

```
    try {
```

```
        FileReader fileReader =  
            new FileReader(fileName);
```

```
        BufferedReader bufferedReader =  
            new BufferedReader(fileReader);
```

```
        String record;  
        String allRecords = "";
```

```
        while ((record =  
            bufferedReader.readLine()) != null) {
```

```
            String[] data = record.split(",");
```

```
            allRecords +=  
                "Employee ID: " + data[0] +  
                "\nName: " + data[1] +  
                "\nDepartment: " + data[2] +  
                "\nSalary: " + data[3] +  
                "\n-----\n";
```

```
        }
```

```
        bufferedReader.close();
```

```
        if (allRecords.equals("")) {
```

```
        outputArea.setText(  
            "No employee records available."  
        );  
  
    } else {  
  
        outputArea.setText(allRecords);  
    }  
  
}  
  
catch (FileNotFoundException ex) {  
  
    outputArea.setText(  
        "Employee file is not available."  
    );  
}  
  
catch (IOException ex) {  
  
    outputArea.setText(  
        "Error while reading employee records."  
    );  
}  
}  
  
public void searchEmployee() {  
  
    String searchId = idField.getText();  
  
    if (searchId.equals("")) {
```

```
        outputArea.setText(  
            "Enter Employee ID to search."  
        );  
  
        return;  
    }  
  
    boolean found = false;  
  
    try {  
  
        FileReader fileReader =  
            new FileReader(fileName);  
  
        BufferedReader bufferedReader =  
            new BufferedReader(fileReader);  
  
        String record;  
  
        while ((record =  
            bufferedReader.readLine()) != null) {  
  
            String[] data = record.split(",");  
  
            if (data[0].equals(searchId)) {  
  
                outputArea.setText(  
                    "Employee Found\n\n" +  
                    "Employee ID: " + data[0] +  
                    "\nName: " + data[1] +  
                    "\nDepartment: " + data[2] +  
                    "\nSalary: " + data[3]  
                );  
                found = true;  
            }  
        }  
    }  
}
```



```
        );

        found = true;

        break;
    }
}

bufferedReader.close();

if (!found) {

    outputArea.setText(
        "Employee record not found."
    );
}

}

catch (FileNotFoundException ex) {

    outputArea.setText(
        "Employee file is not available."
    );
}

catch (IOException ex) {

    outputArea.setText(
        "Error while searching employee record."
    );
}
```

```
    }  
}
```

HTML CODE:

```
<html>  
  
<head>  
  
    <title>Employee Record Management</title>  
</head>  
  
<body>  
  
    <applet  
        code="EmployeeRecordManagement.class"  
        width="600"  
        height="500">  
    </applet>  
  
</body>  
</html>
```

EMPLOYEE POLICY:

```
grant {  
    permission java.io.FilePermission  
        "C:\\javaApplet\\EmployeeRecordManagement\\-",  
        "read,write,delete";  
};
```